KFUPM Global - Intelligent Planet Hackathon 2025 Title: Smarter AI for Greener Earth

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Problem Description

Modern data centres are critical infrastructure supporting cloud services, enterprise systems, and increasingly AI workloads. These data centers consume massive amounts of electricity and water for computation and cooling, resulting in high carbon emissions and environmental stress. While, AI workloads significantly amplify this impact due to intensive computation, most data centers-AI-enabled or not-lack unified, real-time visibility into their environmental footprint**.** Energy usage, water consumption, and carbon emissions are often tracked in silos or reported retrospectively, limiting the ability to optimize operations proactively.

Additionally, stakeholders such as data center operators, AI engineers, sustainability teams, and regulators lack a shared platform to monitor performance, set measurable sustainability goals, and validate progress. Therefore, there is a critical need for an intelligent, data-driven system that works across all data centers**,** regardless of AI usage, while providing deeper insights for AI-intensive workloads—enabling measurable reductions in energy use, water consumption, and carbon emissions.

Research & Supporting Evidence

AI has a significant environmental problem. Data centers housing AI servers produce electronic waste, consume large amounts of water, rely on critical minerals often mined unsustainably, and require massive electricity, spurring greenhouse gas emissions (UNEP, 2025; Navigating New Horizons, 2025). “There is still much we don’t know about the environmental impact of AI but some of the data we do have is concerning,” said Golestan (Sally) Radwan, Chief Digital Officer, UNEP (UNEP, 2025).

Examples of AI’s resource use:

- Making a 2 kg computer requires 800 kg of raw materials; AI microchips need rare earth elements often mined destructively (Navigating New Horizons, 2025).
- Data centers produce hazardous electronic waste containing substances like mercury and lead (UNEP, 2025).
- AI infrastructure may consume six times more water than Denmark, and a single ChatGPT request uses ten times the electricity of a Google Search (WEF, 2025).
- In Ireland, AI data centers could account for nearly 35% of national energy use by 2026 (OECD, 2022).
- AI-powered self-driving cars may encourage more driving over cycling or public transit, and AI can generate climate misinformation (JISC, 2025).

Uneven Distribution of AI’s Environmental Impacts:

Training a single AI model can consume thousands of MWh and emit hundreds of tons of carbon, roughly the annual emissions of hundreds of US households (HBR, 2024). AI model training can evaporate vast amounts of freshwater for cooling, worsening water

scarcity (HBR, 2024). Global AI energy demand may increase tenfold, exceeding Belgium's annual electricity use, while US data centers could reach 6% of national electricity consumption by 2026 (IEA, 2025). Vulnerable communities disproportionately bear these environmental impacts, highlighting inequality (HBR, 2024). Geographical load balancing can reduce carbon intensity by dynamically aligning energy demand with grid conditions (HBR, 2024; Google, 2025).

How AI use impacts the environment and what you can do about it?

AI can have a big impact on the environment because running AI models and data centers uses a lot of electricity and water. This leads to more carbon emissions and can strain natural resources. It is important to monitor and optimize AI usage to reduce its environmental footprint.

Identifying Users & Stakeholders

- **Primary Users / Direct Stakeholders**

Stakeholder	Role / Why They Matter	Supporting Evidence / Quote
AI/ML Engineers C Developers	Design, train, and deploy AI models that require intensive computation, resulting in high energy and water consumption; need real-time insights to optimize model training, inference, and scheduling.	Morrison et al. (2025) show that training large AI models consumes millions of liters of water and emits hundreds of tons of CO_2.
Cloud Infrastructure Teams	Allocate compute resources, manage regional deployments, and balance workloads across data centers; play a key role in energy efficiency and carbon-aware scheduling.	IEA (2025) highlights that cloud infrastructure decisions strongly influence data-center energy demand and emissions.
Data Center Operators	Manage servers, cooling systems, and power infrastructure; responsible for implementing operational changes to reduce energy use, water consumption, and emissions.	Qi et al. (2023) demonstrate that intelligent data-center management significantly reduces energy, carbon, and wastewater impact.
Facilities & Cooling Engineers	Design and operate cooling systems that account for a major share of water and electricity usage in data centers.	Uptime Institute (2024) reports that cooling alone can account for up to 40% of a data center's total energy consumption.
Sustainability / ESG Teams	Track environmental KPIs, ensure compliance with sustainability goals, and report emissions	Gupta & Parmar (2024) emphasize the importance of

	and water usage to regulators and stakeholders.	AI-driven sustainability metrics for ESG reporting and governance.

- **Secondary Users / Stakeholders**

Stakeholder	Role / Why They Matter	Supporting Evidence / Quote
Sustainability / ESG Teams	Monitor environmental KPIs (energy, water, carbon), ensure alignment with corporate sustainability goals, and prepare ESG and regulatory reports.	Gupta & Parmar (2024) highlight that AI-enabled monitoring is essential for transparent ESG reporting and sustainable data governance.
Policy Makers / Regulators	Define environmental standards, carbon limits, and water-usage regulations for data centers; rely on accurate data to enforce compliance.	OECD (2022) emphasizes the need for standardized measurement of AI and data-center environmental footprints for policy development.
Utility Providers (Electricity & Water)	Supply energy and water to data centers; need demand forecasts to manage grid stability and water stress.	IEA (2025) reports that rising data-center demand significantly affects national electricity grids and resource planning.
Enterprise & Cloud Customers	Organizations hosting applications and AI workloads in data centers; increasingly demand transparency on carbon footprint and sustainability.	World Economic Forum (2025) notes growing pressure from customers for environmentally responsible digital services.
Researchers & Academics	Study environmental impacts of AI and data centers; use aggregated data to develop optimization models and sustainability frameworks.	Morrison et al. (2025) stress the lack of accessible, standardized data for holistic AI environmental impact research.

- **Indirect Stakeholders / General Public**

Stakeholder	Role / Why They Matter	Supporting Evidence / Quote

Local Communities	Live near data centers and are affected by high water consumption, land use, and increased electricity demand.	UNEP (2025) warns that data-center expansion can exacerbate local water scarcity and environmental stress.
Society / General Public	Experience indirect impacts such as climate change, higher energy costs, and resource depletion caused by large-scale data-center operations.	HBR (2024) notes that AI-related emissions contribute to broader climate risks affecting society at large.
Environmental NGOs & Advocacy Groups	Advocate for sustainable technology practices and influence policy and corporate accountability.	UNEP (2025) highlights the role of civil society in pushing for greener AI and digital infrastructure.
Future Generations	Bear long-term consequences of today's energy consumption, emissions, and resource depletion.	IPCC-aligned research cited by IEA (2025) stresses that unchecked emissions from digital infrastructure threaten long-term climate stability.

Solution Concept

We propose **EcoMinds Monitor**, a real-time environmental footprint monitoring and optimization system for data centers**,** with advanced analytics for AI-intensive workloads**.** The system continuously tracks energy consumption, water usage, and carbon emissions across compute infrastructure, cooling systems, and workloads—whether AI-based or traditional. By integrating data from data center infrastructure, workload management platforms, and external environmental sources (such as grid carbon intensity and regional water stress), the platform provides actionable insights to reduce environmental impact. AI/ML engineers receive workload-level recommendations to optimize training and inference, while data center operators are guided on cooling efficiency, capacity planning, and resource allocation.

Sustainability and ESG teams benefit from unified dashboards that translate technical metrics into measurable sustainability KPIs**,** supporting regulatory compliance and transparent reporting. Policymakers and regulators can leverage aggregated insights to inform environmental standards and digital infrastructure planning.By enabling carbon-aware workload scheduling, predictive resource forecasting, and continuous performance monitoring, EcoMinds Monitor transforms sustainability goals into measurable actions. The system addresses critical concerns highlighted by Morrison et al. (2025), SHIELD (Qi et al., 2023), and UNEP (2025), ensuring that the growth of AI and digital infrastructure is environmentally responsible and sustainable. **While the platform supports all data centers, AI workloads receive deeper analysis due to their higher computational intensity and environmental impact.**

Google Technology Usage

- **Data Ingestion & Processing – Firebase & Cloud Functions**: Collect real-time energy and water usage, clean, validate, and process data for storage.
- **Data Analysis & Warehousing – BigQuery**: Store and analyze data; calculate real-time emissions; historical trend analysis; predictive modeling with BigQuery ML.
- **AI-Powered Insights – Vertex AI**: Generate recommendations for hardware configurations, workload scheduling, and carbon/water reduction.
- **Macro-Environmental Impact Assessment – Google Earth Engine**: Correlate data center metrics with regional water stress and climate datasets.

Data Sources & Impact Calculations

- EcoMinds Monitor integrates data from multiple internal and external sources to ensure accurate environmental impact assessment. Internal data is collected from data center infrastructure management systems, including server-level energy meters, cooling system water usage sensors, and workload orchestration platforms that provide CPU/GPU utilization metrics.
- External environmental data is sourced from trusted public datasets, such as grid-level carbon intensity data from the International Energy Agency (IEA) and Electricity Maps, and regional water stress indicators from the World Resources Institute (WRI) Aqueduct and Google Earth Engine.
- Carbon emissions are calculated using standardized formulas by multiplying energy consumption (kWh) by region-specific grid carbon intensity (kg CO_2/kWh). Water efficiency metrics are computed using industry-recognized indicators such as Water Usage Effectiveness (WUE). These calculations follow methodologies recommended by the IEA, OECD, and Uptime Institute, ensuring transparency, accuracy, and comparability across regions.

AI Features

- **Real-time Monitoring:** Tracks energy, water, and carbon continuously.
- **Predictive Insights:** Forecasts resource consumption and environmental impact.
- **Optimization Recommendations:** Suggests strategies to reduce energy/water use.
- **Intelligent Load Distribution:** Balances workloads across data centers for lower carbon intensity.
- **Alerts & Risk Management:** Notifies users of excessive usage or environmental risks.
- **Reporting & Visualization:** Interactive dashboards for engineers, ESG teams, policymakers.
- **Compliance Support:** Ensures alignment with sustainability goals and regulations.

Innovation

- AI-Powered Energy Optimizer: Vertex AI analyzes workloads to recommend optimal schedules and regions.
- Eco Score Rating System: Sustainability score for each AI workload based on energy, carbon, and water.
- Predictive Resource Forecasting: ML predicts future energy and water demands.
- Gemini-Powered Smart Recommendations: Natural-language tips for greener operations.
- Unified Sustainability Dashboard: Combines workload data, metrics, and optimization insights.

Its Impact and Alignment

EcoMinds Monitor directly contributes to global and national sustainability goals by helping AI-driven organizations measure, monitor, and reduce their environmental footprint. It reduces energy and water consumption in AI workloads, cutting operational costs while lowering carbon emissions. By identifying high-impact regions and enabling load balancing toward renewable-powered zones, it promotes more efficient data center management.

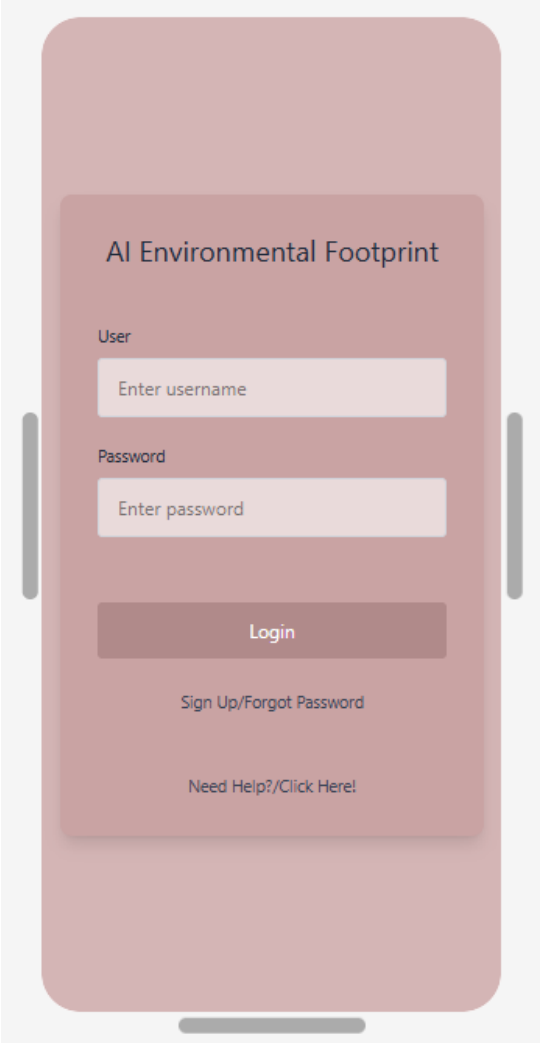
The platform empowers businesses to track their CO_2 footprint and progress toward net-zero commitments. It aligns with Saudi Vision 2030, supporting sustainable digital transformation, clean energy adoption, and responsible technology growth. It also advances the UN Sustainable Development Goals:

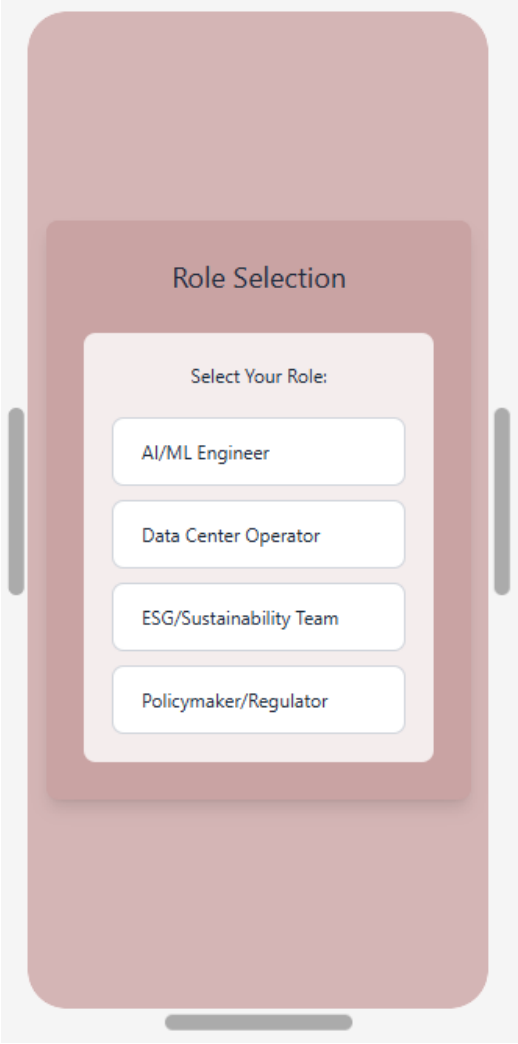
- Goal 7 – Affordable and Clean Energy: Encourages use of renewable-powered data center regions.
- Goal 9 – Industry, Innovation, and Infrastructure: Integrates AI innovation into sustainable infrastructure.
- Goal 12 – Responsible Consumption and Production: Promotes energy optimization and waste reduction.
- Goal 13 – Climate Action: Lowers carbon emissions from large-scale AI operations.

By combining these features and aligning with strategic goals, EcoMinds Monitor transforms environmental responsibility into a measurable, achievable, and intelligent process, positioning itself as a global standard for green AI operations.

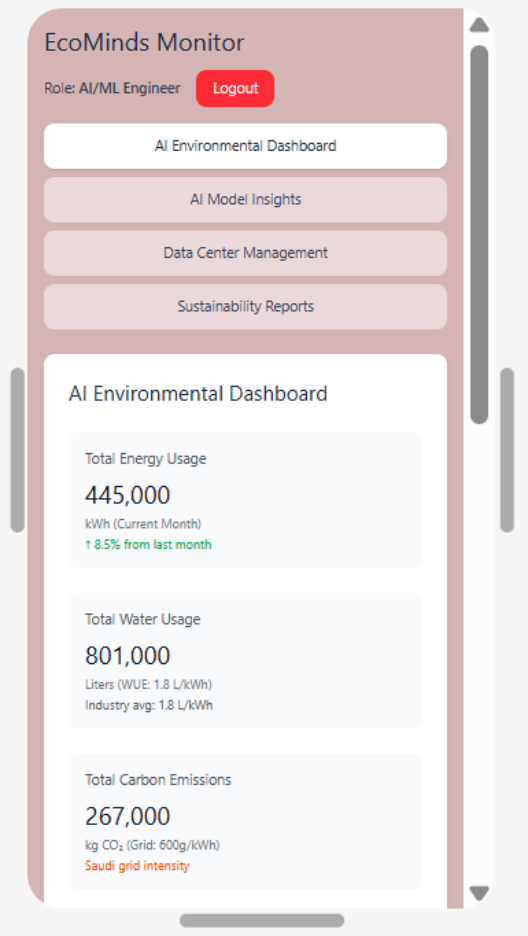
UI Mockup and System Architecture

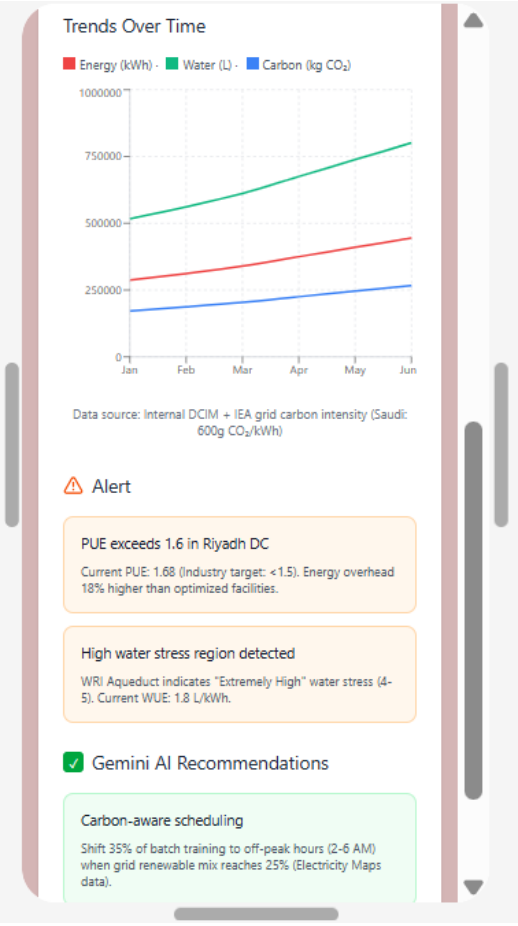
These UI mockups, created using Figma, illustrate key screens for real-time monitoring, model- specific insights, data center management, and reporting—demonstrating how the system functions, is feasible to implement, and supports sustainability goals by reducing energy, water, and carbon footprints.



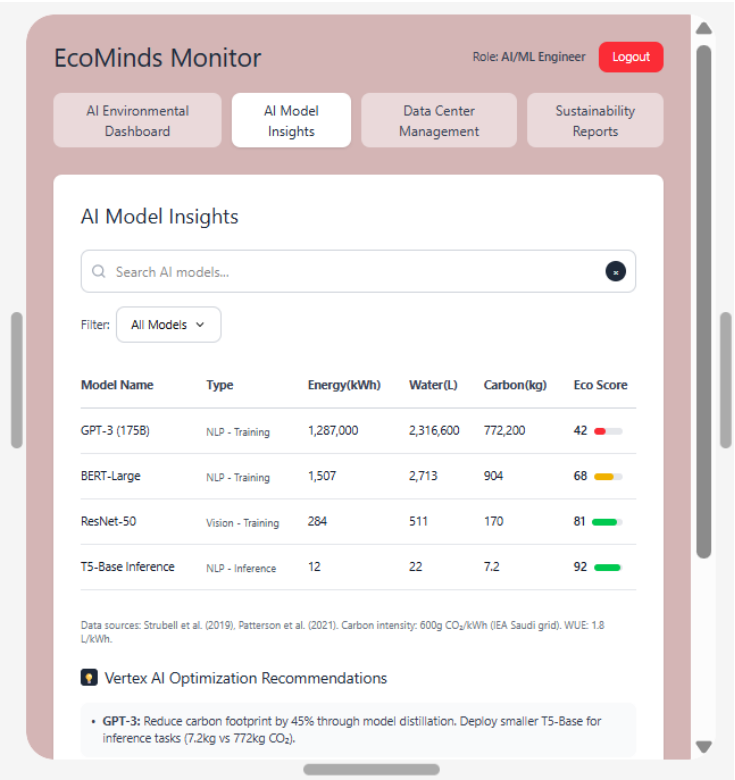


AI Environmental Dashboards

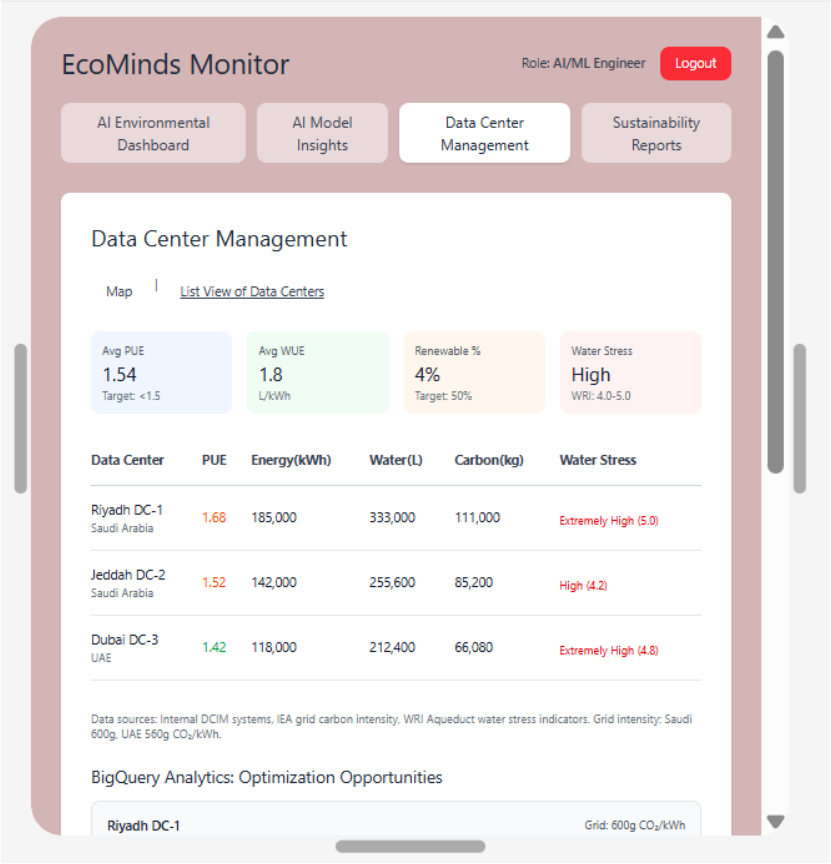




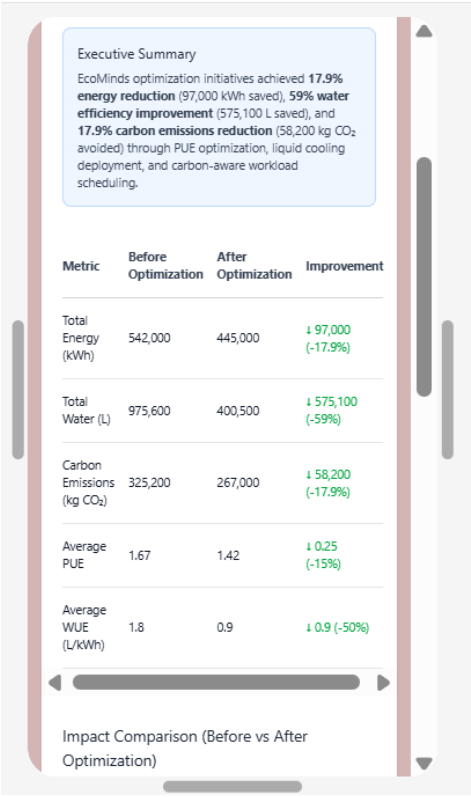
AI Model Insights



Data Center Management



Sustainability Reports

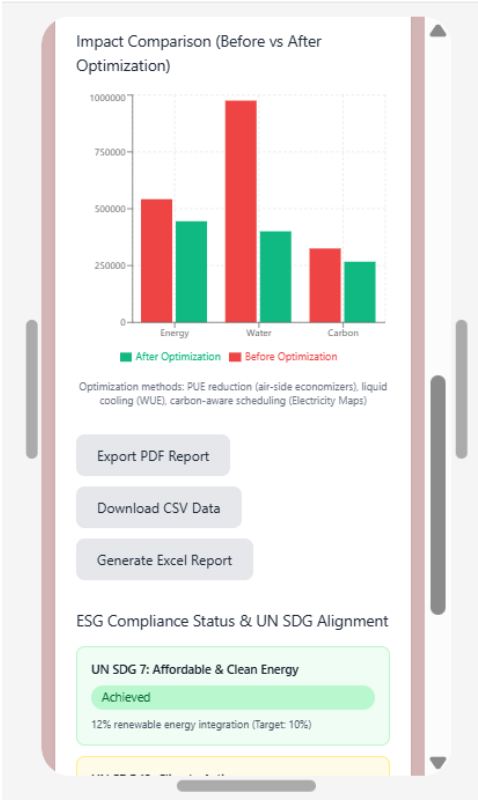


Executive Summary

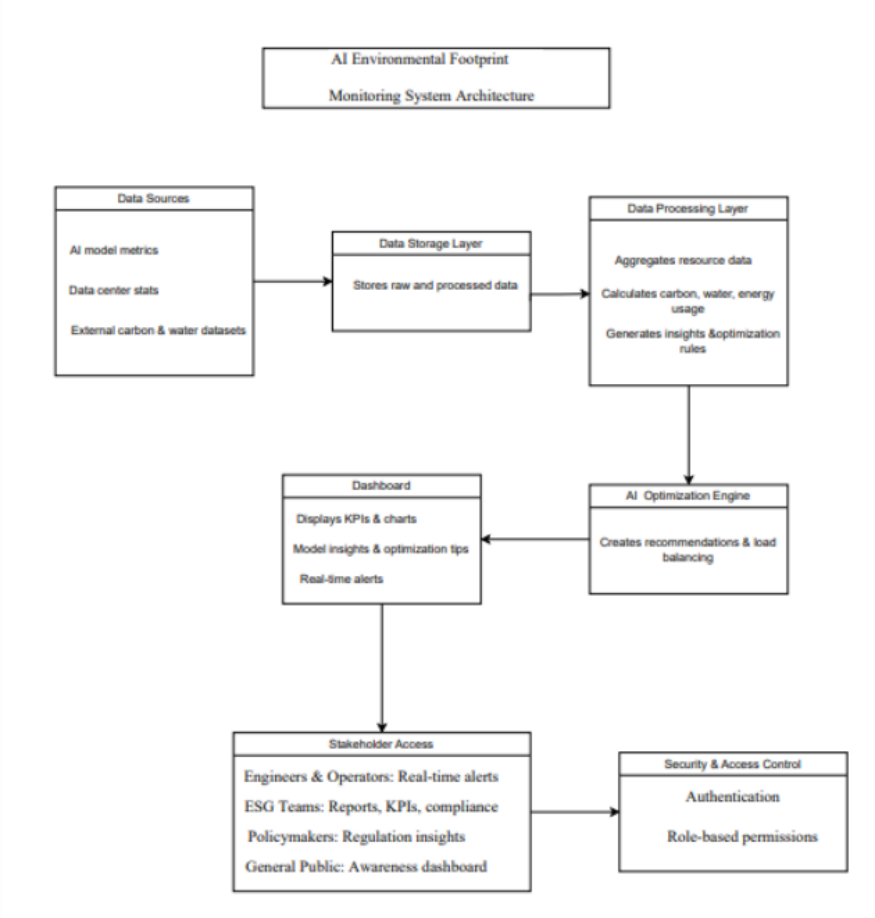
EcoMinds optimization initiatives achieved **17.9% energy reduction** (97,000 kWh saved), **59% water efficiency improvement** (575,100 L saved), and **17.9% carbon emissions reduction** (58,200 kg CO_2 avoided) through PUE optimization, liquid cooling deployment, and carbon-aware workload scheduling.

Metric	Before Optimization	After Optimization	Improvement
Total Energy (kWh)	542,000	445,000	↓ 97,000 (-17.9%)
Total Water (L)	975,600	400,500	↓ 575,100 (-59%)
Carbon Emissions (kg CO_2)	325,200	267,000	↓ 58,200 (-17.9%)
Average PUE	1.67	1.42	↓ 0.25 (-15%)
Average WUE (L/kWh)	1.8	0.9	↓ 0.9 (-50%)

Impact Comparison (Before vs After Optimization)



System Architecture Diagram



The system collects data from AI models, data centers, and external sources, processes it to calculate environmental impact, and generates actionable insights. Dashboards, reports, model insights, alerts, and data center management reflect stakeholder access, reporting, and optional security layers, as shown in the draw.io architecture diagram.

Future Scope

EcoMinds Monitor is an innovative platform that transforms AI into part of the solution for AI's environmental footprint. Future enhancements include extending the AI-Powered Energy Optimizer to multi-cloud environments for global workload efficiency, integrating regional renewable energy metrics to guide data center operations, and refining the Eco Score rating system to incorporate full hardware lifecycle and supply chain impacts. Predictive analytics will forecast future energy, water, and carbon demand for proactive sustainability planning. The Gemini-powered smart recommendations can evolve to provide personalized operational guidance in real time, helping organizations align with green policies. The Unified Sustainability Dashboard will aggregate AI workload, environmental metrics, and optimization insights into a single interface for engineers, ESG teams, and policymakers. By combining Google technologies like Vertex AI, BigQuery, Gemini, and Maps Platform, EcoMinds Monitor positions itself as a scalable, globally adoptable standard for sustainable AI operations, fully aligned with Saudi Vision 2030 and UN SDGs 7, 9, 12, and 13.

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